

Big Data: End-to-End Linear Regression Pipeline

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Introduction

When data became cheaper and easier to store, it created a new problem. Analysts no longer suffered from data scarcity, but handling the massive datasets became more cumbersome. Luckily, many solutions have been developed to address this problem, and this project demonstrates how multiple solutions can be used within a single big data pipeline.

Dataset

Since the project focuses on the pipeline itself and was deployed on Google Cloud's free-trial virtual machine, a small synthetic dataset was used to reduce compute costs, while still demonstrating the basic concepts. The dataset was generated by AI using ChatGPT and contains the following variables: student_id, sleep_hours, gpa, and bedtime_category (early, moderate, or late). The goal of the pipeline is to simulate a big data approach to predicting students' GPAs using sleep-related features.

Pipeline Overview

The pipeline will use six software components, each designed to serve a distinct purpose within the big data pipeline: NiFi, HDFS, Apache Hive, Apache Spark, HBase, and YARN. NiFi will be used for data ingestion, HDFS for data storage, YARN for resource management, Hive for data warehousing, HBase for NoSQL storage, and Spark for data processing, querying, and machine learning. The data will be ingested via NiFi, loaded into HDFS, loaded into a Hive table, analyzed in Spark using SQL and MLlib, and the final evaluation metrics will be written to HBase for easy querying.

Issues Encountered

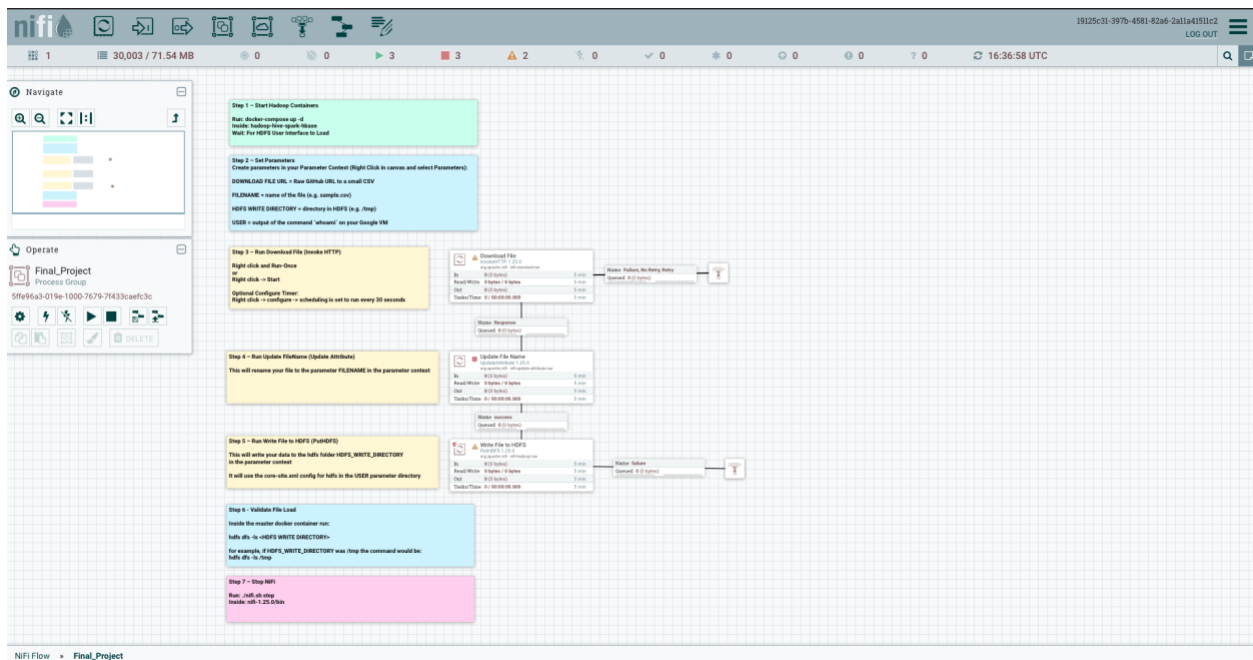
While the pipeline was successful end-to-end, an error occurred when I initially loaded the data into HDFS via NiFi. The process showed a single error within the queue. On further investigation, it was discovered that the error occurred because the file had already been loaded into HDFS. An extra click must

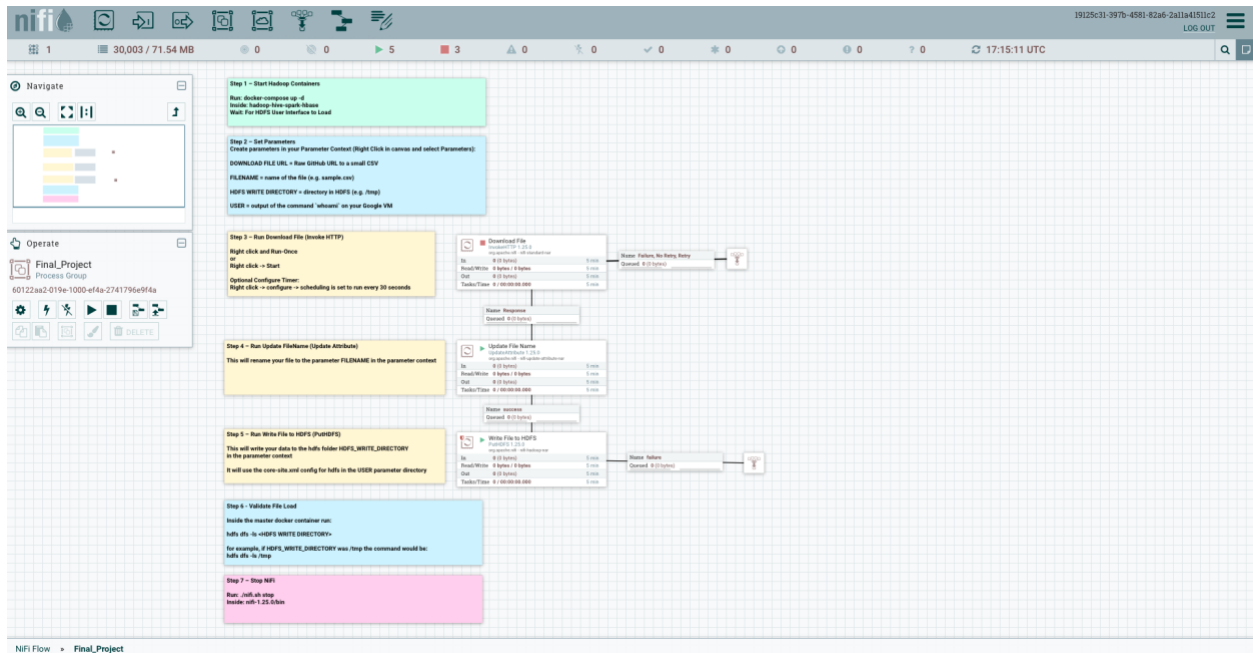
have caused the data to be ingested more than once, causing the error. It was easy to remove the error from the queue, and the verification command output proved that the file had been loaded to HDFS without issue.

Screenshots & Code

NiFi Data Ingestion into HDFS

This NiFi flow uses InvokeHTTP, UpdateAttribute and PutHDFS to ingest the raw github file into HDFS. InvokeHTTP retrieves the file from the GitHub raw URL that was entered when defining the parameters, and brings it into NiFi as a FlowFile. UpdateAttribute modifies the filename to the specified name, also established in the parameters, and PutHDFS writes the FlowFile to the targeted HDFS directory. The file was successfully written into the HDFS directory, as confirmed by the `hdfs dfs -ls` command output.





```
bash-5.0# hdfs dfs -ls /tmp
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2026-05-25 17:06:55,696 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
using builtin-java classes where applicable
Found 2 items
drwx-wx-wx - root supergroup 0 2026-05-25 16:47 /tmp/hive
-rw-r--r-- 3 charlottealley supergroup 918 2026-05-25 16:51 /tmp/simulated_student_sleep_gpa.csv
```

Hive Managed Table

The schema was designed to mimic the structure of the simulated student sleep data CSV file. INT was used for the student_id file because the column uses whole numbers. The sleep_hours and gpa columns were set to DOUBLE because they contained decimal values, similar to using the float data type in Python. The bedtime_category column uses STRING because it contains categorical text values. The table uses comma-separated values formatting because the original source is a CSV file, and the header row is skipped to avoid any errors with the inputted values.

```
CREATE TABLE simulated_student_sleep_gpa (  
  
  > student_id INT,  
  
  > sleep_hours DOUBLE,  
  
  > gpa DOUBLE,  
  
  > bedtime_category STRING  
  
  > )  
  
  > ROW FORMAT DELIMITED  
  
  > FIELDS TERMINATED BY ','  
  
  > STORED AS TEXTFILE  
  
  > TBLPROPERTIES ("skip.header.line.count"="1");
```

```
hive> LOAD DATA INPATH '/tmp/simulated_student_sleep_gpa.csv'  
> INTO TABLE simulated_student_sleep_gpa;  
Loading data to table default.simulated_student_sleep_gpa  
OK  
Time taken: 1.183 seconds  
hive> SELECT * FROM simulated_student_sleep_gpa LIMIT 10;  
OK  
1      6.2      3.1      late  
2      7.5      3.6      moderate  
3      5.8      2.9      late  
4      8.1      3.8      early  
5      6.9      3.3      moderate  
6      7.2      3.5      moderate  
7      5.4      2.7      late  
8      8.4      3.9      early  
9      6.7      3.2      moderate  
10     7.9      3.7      early  
Time taken: 3.474 seconds, Fetched: 10 row(s)
```

```
hive> SELECT
>   bedtime_category,
>   COUNT(*) AS student_count,
>   ROUND(AVG(sleep_hours), 2) AS avg_sleep_hours,
>   ROUND(AVG(gpa), 2) AS avg_gpa
> FROM simulated_student_sleep_gpa
> GROUP BY bedtime_category;
Query ID = root_20260525173929_3476962a-607b-441c-8dfa-ab9502d80fe0
Total jobs = 1
Launching Job 1 out of 1
Tez session was closed. Reopening...
2026-05-25 17:39:31,971 INFO [2ce88fce-e1fd-45eb-98d8-e07631ced679 main] client.RMProxy: Connecting to ResourceManager at master/172.28.1.1:8032
Session re-established.
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1779727557601_0004)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1		container	SUCCEEDED	1	1	0	0	0	0
Reducer 2		container	SUCCEEDED	1	1	0	0	0	0

```
VERTICES: 02/02 [=====] 100% ELAPSED TIME: 7.78 s
OK
early 15 8.13 3.83
late 18 5.82 2.82
moderate 17 7.06 3.39
Time taken: 17.686 seconds, Fetched: 3 row(s)
```

Environment Setup

The environment setup steps are necessary because the pipeline uses multiple big data tools, each serving its own purpose. By installing HappyBase, Python can connect to HBase through the Thrift interface and use its supporting packages (such as numpy). Together, these steps allow the data to move from file-based storage to queryable big data systems.

```
root@ec2-54-186-121-100:~/bigdata# docker-compose exec worker1 bash
bash-5.0# pip install numpy happybase
Collecting numpy
  Downloading numpy-1.21.6.zip (18.3 MB)
    | 18.3 MB 5.7 MB/s
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing wheel metadata ... done
Collecting happybase
  Downloading happybase-1.3.0-py2.py3-none-any.whl (26 kB)
Collecting importlib-resources
  Downloading importlib-resources-5.12.0-py3-none-any.whl (36 kB)
Collecting six
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Collecting thriftypy2==0.4
  Downloading thriftypy2-0.4.0.tar.gz (996 kB)
    | 996 kB 37.3 MB/s
  Installing build dependencies ... done
  WARNING: Missing build requirements in pyproject.toml for thriftypy2==0.4 from https://files.pythonhosted.org/packages/cc/7c/bc838fb68b6b5474181197d6286f6c1839c47739816587a748ac8541/thriftypy2-0.4.0.tar.gz#sha256=621c263f99274d51e9a1deecf381845f1488d497d4fed682db
  WARNING: The project does not specify a build backend, and pip cannot fall back to setuptools without 'wheel'.
  Getting requirements to build wheel ... done
  Installing backend dependencies ... done
  Preparing wheel metadata ... done
Collecting ripp==0.1.0; python_version < '3.10'
  Downloading ripp-0.1.0-py3-none-any.whl (6.8 kB)
Collecting typing-extensions==3.7.4; python_version < '3.8'
  Downloading typing-extensions-4.7.1-py3-none-any.whl (23 kB)
Collecting ply==4.0.2014
  Downloading ply-4.0-py2.py3-none-any.whl (109 kB)
    | 109 kB 6.3 MB/s
building wheels for collected packages: numpy, thriftypy2
  Building wheel for numpy (PEP 517) ... done
  Created wheel for numpy: filename=numpy-1.21.6-cp37-cp37m-linux_x86_64.whl size=16644471 sha256=f3c4e1513bba16f5da7786427afbea44d5d598a64a54182edcbba613cde
  Stored in directory: /root/.cache/pip/wheels/6e/7e/9e/8de42c7f2a93994a76dec9c3fb78f6b4c3bc94eb3bea
  Building wheel for thriftypy2 (PEP 517) ... done
  Created wheel for thriftypy2: filename=thriftypy2-0.4.0-cp37-cp37m-linux_x86_64.whl size=1797951 sha256=2893d18b57d14f9017d5534273e484cf28db9e8f6f32f5d7f1a54ca115baee
  Stored in directory: /root/.cache/pip/wheels/cv/c1/63/8278f96821c96cd3b0397273bde4bc189489598338d6d3
Successfully built numpy thriftypy2
Installing collected packages: numpy, ripp, importlib-resources, six, typing-extensions, ply, thriftypy2, happybase
Successfully installed happybase-1.3.0 importlib-resources-5.12.0 numpy-1.21.6 ply-4.0 ripp-0.1.0 six-1.17.0 thriftypy2-0.4.0 typing-extensions-4.7.1 ripp-0.1.0
bash-5.0# exit
```

```

[sharlott@llydsdcs02-bigdata:~/dcs02-infra/bellevue-bigdata/hadoop-niwe-spark-thrift$ docker-compose exec worker2 bash
bash-5.0# pip3 install numpy happybase
Collecting numpy
  Downloading numpy-1.21.4.zip (10.3 MB)
    Installing build dependencies ... done
    Getting requirements to build wheel ... done
    Preparing wheel metadata ... done
Collecting happybase
  Downloading happybase-1.3.0-py2.py3-none-any.whl (26 kB)
Collecting importlib-resources
  Downloading importlib_resources-5.12.0-py3-none-any.whl (36 kB)
Collecting six
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Collecting thriftpy2==0.4
  Downloading thriftpy2==0.4.0.tar.gz (996 kB)
    Installing build dependencies ... done
    WARNING: Missing build requirements in pyproject.toml for thriftpy2==0.4 from https://files.pythonhosted.org/packages/c0/7c/b038fb6868b6b647a1811976d286a6fbc1829c4f739816587a748ac68541/thriftpy2-0.4.0.tar.gz#sha256=421c263f9927ad01a9a1deecf381845f1888d497bdaFed682db615512290f6c (from happybase).
    WARNING: The project does not specify a build backend, and pip cannot fall back to setuptools without 'wheel'.
    Getting requirements to build wheel ... done
    Installing backend dependencies ... done
    Preparing wheel metadata ... done
Collecting zipp>=3.6; python_version < "3.10"
  Downloading zipp-3.10.0-py3-none-any.whl (6.8 kB)
Collecting typing-extensions>=3.7.4; python_version < "3.8"
  Downloading typing_extensions-4.7.1-py3-none-any.whl (33 kB)
Collecting ply>=4.0; >=3.4
  Downloading ply-3.11-py3-none-any.whl (49 kB)
    Building wheel for numpy (PEP 517) ... done
    Building wheel for happybase (PEP 517) ... done
    Created wheel for numpy: filename=numpy-1.21.4-cp37-cp37m-linux_x86_64.whl size=56643880 sha256=6e7378b18e4451826248c098a6f83a1837f6413a58688de793f63b678f9c5
    Stored in directory: /root/.cache/pip/wheels/4e/7a/7e/8fde842ccff7263994874dac9c3bf67878b444c3bd94eb386a
    Building wheel for thriftpy2 (PEP 517) ... done
    Created wheel for thriftpy2: filename=thriftpy2-0.4.0-cp37-cp37m-linux_x86_64.whl size=1479984 sha256=35122e67999c19c49dc93bb7f88f42e94ae78a989f13a75dc833bc189158
    Stored in directory: /root/.cache/pip/wheels/c0/e1/d3/38778f9881c96dd3ed3b392293b6e0c18f9e9588338db63
    Successfully built numpy thriftpy2
Installing collected packages: numpy, zipp, importlib-resources, six, typing-extensions, ply, thriftpy2, happybase
Successfully installed happybase-1.3.0 importlib-resources-5.12.0 numpy-1.21.4 ply-3.11 six-1.17.0 thriftpy2-0.4.0 typing-extensions-4.7.1 zipp-3.10.0

```

```

[sharlott@llydsdcs02-bigdata:~/dcs02-infra/bellevue-bigdata/hadoop-niwe-spark-thrift$ docker-compose exec master bash
bash-5.0# pip3 install numpy happybase
Collecting numpy
  Downloading numpy-1.21.4.zip (10.3 MB)
    Installing build dependencies ... done
    Getting requirements to build wheel ... done
    Preparing wheel metadata ... done
Collecting happybase
  Downloading happybase-1.3.0-py2.py3-none-any.whl (26 kB)
Collecting importlib-resources
  Downloading importlib_resources-5.12.0-py3-none-any.whl (36 kB)
Collecting six
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Collecting thriftpy2==0.4
  Downloading thriftpy2==0.4.0.tar.gz (996 kB)
    Installing build dependencies ... done
    WARNING: Missing build requirements in pyproject.toml for thriftpy2==0.4 from https://files.pythonhosted.org/packages/c0/7c/b038fb6868b6b647a1811976d286a6fbc1829c4f739816587a748ac68541/thriftpy2-0.4.0.tar.gz#sha256=421c263f9927ad01a9a1deecf381845f1888d497bdaFed682db615512290f6c (from happybase).
    WARNING: The project does not specify a build backend, and pip cannot fall back to setuptools without 'wheel'.
    Getting requirements to build wheel ... done
    Installing backend dependencies ... done
    Preparing wheel metadata ... done
Collecting zipp>=3.6; python_version < "3.10"
  Downloading zipp-3.10.0-py3-none-any.whl (6.8 kB)
Collecting typing-extensions>=3.7.4; python_version < "3.8"
  Downloading typing_extensions-4.7.1-py3-none-any.whl (33 kB)
Collecting ply>=4.0; >=3.4
  Downloading ply-3.11-py3-none-any.whl (49 kB)
    Building wheel for numpy (PEP 517) ... done
    Building wheel for happybase (PEP 517) ... done
    Created wheel for numpy: filename=numpy-1.21.4-cp37-cp37m-linux_x86_64.whl size=56644278 sha256=cf221a7af58182c4857686985e55647a75363952d42c0b7e6421e672e6d80f9a
    Stored in directory: /root/.cache/pip/wheels/4e/7a/7e/8fde842ccff7263994874dac9c3bf67878b444c3bd94eb386a
    Building wheel for thriftpy2 (PEP 517) ... done
    Created wheel for thriftpy2: filename=thriftpy2-0.4.0-cp37-cp37m-linux_x86_64.whl size=1479773 sha256=1ee88af34e8cd25c9e88b95d23c8f47f99203a7b2848e244d1a532293aca3845
    Stored in directory: /root/.cache/pip/wheels/c0/e1/d3/38778f9881c96dd3ed3b392293b6e0c18f9e9588338db63
    Successfully built numpy thriftpy2
Installing collected packages: numpy, zipp, importlib-resources, six, typing-extensions, ply, thriftpy2, happybase
Successfully installed happybase-1.3.0 importlib-resources-5.12.0 numpy-1.21.4 ply-3.11 six-1.17.0 thriftpy2-0.4.0 typing-extensions-4.7.1 zipp-3.10.0
bash-5.0#

```

```

bash-5.0# nohup hbase thrift start &
[3] 7763
bash-5.0# nohup: ignoring input and appending output to 'nohup.out'
jps
528 NameNode
2688 HistoryServer
1089 RunJar
1090 RunJar
1060 JobHistoryServer
2741 HMaster
662 SecondaryNameNode
7894 Jps
762 ResourceManager
7626 ThriftServer
2684 Master
[3]+  Exit 1                  nohup hbase thrift start

```

HBase Table Creation

The HBase row key was designed with the `student_id` in mind, because each student has a unique identifier in the synthetic dataset. This design makes it easy to look up each individual record. The table uses a single-column family to store each column, because all the fields describe the same student record and are likely to be accessed together. This keeps the HBase design simple while still supporting quick row-based lookups.

```
hbase(main):001:0> create 'my_table', 'cf'
Created table my_table
Took 1.8855 seconds
```

```
hbase(main):004:0> scan 'my_table'
ROW          COLUMN+CELL
0 row(s)
Took 0.0663 seconds
```

PySpark ML Code

I chose to run a Linear Regression model because the target variable is both numeric and continuous (GPA). The model used `sleep_hours` and `bedtime_category` as input features to predict GPA. This model was appropriate because the goal was to estimate a continuous outcome rather than to classify students into categories. The RMSE shows the average size of the model's prediction error (0.066), while the R^2 shows how much of the variation in GPA the model explains (.978). These results suggest a very strong fit and that sleep-related variables provide useful predictive information for student GPA (which is not surprising, given that simulated data was used).

```
from pyspark.sql import SparkSession
from pyspark.ml.feature import VectorAssembler, StringIndexer
from pyspark.ml.regression import LinearRegression
import happybase

# Step 1: Create a Spark session
```



```

spark = SparkSession.builder \
    .appName("Student Sleep GPA Prediction") \
    .enableHiveSupport() \
    .getOrCreate()

# Step 2: Load data from student sleep Hive table into Spark df
student_df = spark.sql("""
    SELECT
        student_id,
        sleep_hours,
        gpa,
        bedtime_category
    FROM simulated_student_sleep_gpa
""")

# Step 3: Drop null values
student_df = student_df.na.drop()

# Step 4: Convert bedtime_category into a numeric feature
indexer = StringIndexer(
    inputCol="bedtime_category",
    outputCol="bedtime_category_index",
    handleInvalid="skip"
)

indexed_df = indexer.fit(student_df).transform(student_df)

# Step 5: Assemble features
assembler = VectorAssembler(
    inputCols=["sleep_hours", "bedtime_category_index"],
    outputCol="features",
    handleInvalid="skip"
)

assembled_df = assembler.transform(indexed_df).select("features", "gpa")

# Step 6: Split data into training and testing sets

```

```

train_data, test_data = assembled_df.randomSplit([0.7, 0.3], seed=42)

# Step 7: Train Linear Regression model
lr = LinearRegression(labelCol="gpa", featuresCol="features")
lr_model = lr.fit(train_data)

# Step 8: Evaluate model
test_results = lr_model.evaluate(test_data)

print(f"RMSE: {test_results.rootMeanSquaredError}")
print(f"R^2: {test_results.r2}")

# Step 9: Write metrics to HBase
data = [
    ("metrics1", "cf:rmse", str(test_results.rootMeanSquaredError)),
    ("metrics1", "cf:r2", str(test_results.r2)),
]

def write_to_hbase_partition(partition):
    connection = happybase.Connection("master")
    connection.open()
    table = connection.table("my_table")

    for row in partition:
        row_key, column, value = row
        table.put(row_key, {column: value})

    connection.close()

rdd = spark.sparkContext.parallelize(data)
rdd.foreachPartition(write_to_hbase_partition)

# Step 10: Stop Spark
spark.stop()

```

<https://github.com/charlottealley/bigdata>

```

bash-5.0# spark-submit \
> --master yarn \
> --deploy-mode client \
> --name StudentSleepGPA_to_HBase \
> student_sleep_gpa_spark_hbase.py
[16A] Class path contains multiple SLF4J bindings.
[16A] Found binding in [jar:file:/usr/local/hadoop/share/hadoop-common/lib/slf4j-log4j12-1.7.30.jar/org/slf4j/impl/StaticLoggerBinder.class]
[16A] Found binding in [jar:file:/usr/local/hadoop/share/hadoop-common/lib/slf4j-log4j12-1.7.25.jar/org/slf4j/impl/StaticLoggerBinder.class]
[16A] See http://www.slf4j.org/faq.html#multiple_bindings for an explanation.
[16A] Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
0 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
1798 [Thread-5] INFO org.apache.spark.SparkContext - Running Spark version 3.0.0
1800 [Thread-5] INFO org.apache.spark.resource.ResourceUtils - =====
1807 [Thread-5] INFO org.apache.spark.resource.ResourceUtils - Resources for spark.driver:

1808 [Thread-5] INFO org.apache.spark.resource.ResourceUtils - =====
1809 [Thread-5] INFO org.apache.spark.SparkContext - Submitted application: Student Sleep GPA Prediction
1807 [Thread-5] INFO org.apache.spark.SecurityManager - Changing view acls to: root
2807 [Thread-5] INFO org.apache.spark.SecurityManager - Changing modify acls to: root
2807 [Thread-5] INFO org.apache.spark.SecurityManager - Changing view acls groups to:
2807 [Thread-5] INFO org.apache.spark.SecurityManager - Changing modify acls groups to:
2808 [Thread-5] INFO org.apache.spark.SecurityManager - SecurityManager: authentication disabled; ui acls disabled; users with view permissions: Set(root); groups with view permissions: Set(); users with modify permissions: Set(root); groups with modify permissions:
5411
2408 [Thread-5] INFO org.apache.spark.util.Utils - Successfully started service 'sparkDriver' on port 7080.
2409 [Thread-5] INFO org.apache.spark.SparkEnv - Registering MapOutputTracker
2509 [Thread-5] INFO org.apache.spark.SparkEnv - Registering BlockManagerMaster
2637 [Thread-5] INFO org.apache.spark.storage.BlockManagerMasterEndpoint - Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
2638 [Thread-5] INFO org.apache.spark.storage.BlockManagerMasterEndpoint - BlockManagerMasterEndpoint up
2722 [Thread-5] INFO org.apache.spark.SparkEnv - Registering BlockManagerMasterHeartbeat
2768 [Thread-5] INFO org.apache.spark.storage.DiskBlockManager - Created local directory at /tmp/lockmgr-6d5b2316-55c5-4416-ba8a-4a71c88c29b9
2801 [Thread-5] INFO org.apache.spark.storage.memory.MemoryStore - MemoryStore started with capacity 384.0 MiB
2808 [Thread-5] INFO org.apache.spark.SparkEnv - Registering OutputCommitCoordinator
3073 [Thread-5] INFO org.sparkproject.jetty.util.log - Logging initialized @6810ms to org.sparkproject.jetty.util.log.Slf4jLog
3216 [Thread-5] INFO org.sparkproject.jetty.server.Server - jetty-9.4.0-0.M0007; built: 2015-08-29T20:42:08.000Z; gtt: abc35128a6d17e3df052294e43f3a25ce7097; jvm 1.8.0_275-081
3244 [Thread-5] INFO org.sparkproject.jetty.server.Server - Started @594ms
3288 [Thread-5] INFO org.sparkproject.jetty.server.AbstractConnector - Started ServerConnector@93ff6d4a[HTTP/1.1,(http://172.28.1.1:4040)]
3288 [Thread-5] INFO org.apache.spark.util.Utils - Successfully started service 'SparkUI' on port 4040.
3323 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@96d70b6a[/jobs,null,AVAILABLE,@Spark]
3327 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@974ed0b3[/jobs/json,null,AVAILABLE,@Spark]
3328 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@970212ee[/jobs/job,null,AVAILABLE,@Spark]
3332 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@96d40d4f[/jobs/job/json,null,AVAILABLE,@Spark]
3343 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@96d40d4f[/stages,null,AVAILABLE,@Spark]
3364 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@96d52284[/stages/json,null,AVAILABLE,@Spark]
3365 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@929801ed[/stages/rtape,null,AVAILABLE,@Spark]
3367 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@9278a33f[/stages/stage/json,null,AVAILABLE,@Spark]
3367 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@923ba742[/stages/pool,null,AVAILABLE,@Spark]
3368 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@9194d4de[/stages/pool/json,null,AVAILABLE,@Spark]
3383 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@97991f86[/storage,null,AVAILABLE,@Spark]
3383 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@9746d30c[/storage/json,null,AVAILABLE,@Spark]
3384 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@91f655af[/storage/rtdd,null,AVAILABLE,@Spark]
3387 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@974613bb[/storage/rtdd/json,null,AVAILABLE,@Spark]
3389 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@96263f6e[/environment,null,AVAILABLE,@Spark]
3394 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@9640e080[/environment/json,null,AVAILABLE,@Spark]
3394 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@97419598[/executors,null,AVAILABLE,@Spark]
3396 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@9641809f[/executors/json,null,AVAILABLE,@Spark]
3396 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@9336ac78[/executors/threadDump,null,AVAILABLE,@Spark]
3397 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@97719b45[/executors/threadDump/json,null,AVAILABLE,@Spark]
3398 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@96438926[/static,null,AVAILABLE,@Spark]
3398 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@9648e2da[/static,null,AVAILABLE,@Spark]
3398 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@9648e2da[/api,null,AVAILABLE,@Spark]
3399 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@96671737[/jobs/all,null,AVAILABLE,@Spark]
3399 [Thread-5] INFO org.sparkproject.jetty.server.handler.ContextHandler - Started o.s.j.s.ServletContextHandler@974c7444[/stages/stage/all,null,AVAILABLE,@Spark]
3399 [Thread-5] INFO org.apache.spark.ui.SparkUI - Bound SparkUI to 172.28.1.1, and started at http://172.28.1.1:4040
4083 [Thread-5] INFO org.apache.hadoop.yarn.client.NMProxy - Connecting to ResourceManager at master/172.28.1.1:8032
4400 [Thread-5] INFO org.apache.spark.deploy.yarn.Client - Requesting a new application from cluster with 2 NodeManagers
4823 [Thread-5] INFO org.apache.hadoop.conf.Configuration.deprecation - No unit for dfs.client.datanode.restart.timeout(30) assuming SECONDS
5281 [Thread-5] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
5282 [Thread-5] INFO org.apache.hadoop.yarn.util.ResourceUtils - Unable to find 'resource-types.xml'.
5338 [Thread-5] INFO org.apache.spark.deploy.yarn.Client - Verifying our application has not requested more than the maximum memory capability of the cluster (1536 MB per container)
5338 [Thread-5] INFO org.apache.spark.deploy.yarn.Client - Will allocate AM container, with 896 MB memory including 384 MB overhead
5338 [Thread-5] INFO org.apache.spark.deploy.yarn.Client - Setting up container launch context for our AM
5338 [Thread-5] INFO org.apache.spark.deploy.yarn.Client - Setting up the launch environment for our AM container
5355 [Thread-5] INFO org.apache.spark.deploy.yarn.Client - Preparing resources for our AM container

```

```

bash-5.0# grep -i "RMSE|R^2" spark_output.txt
RMSE: 0.06562225809386474
R^2: 0.9778711163548931

```

Spark Submit & Output

```

bash-5.0# spark-submit \
> --master yarn \
> --deploy-mode client \
> --name StudentSleepGPA_to_HBase \
> student_sleep_gpa_spark_hbase.py

```

```

client token: N/A
diagnostics: N/A
ApplicationMaster host: 172.28.1.3
ApplicationMaster RPC port: -1
queue: root.root
start time: 1779736432623
final status: UNDEFINED
tracking URL: http://master:8088/proxy/application-1779727557601_0005/
user: root

```

```

Unit-3-68 yarn logs applicationID application_15797255561_0005
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.18.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/slf4j-log4j-impl-2.10.0.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2022-05-25 19:21:10,383 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2022-05-25 19:21:10,385 INFO client.DumpProxy: Connecting to ResourceManager at master172.24.1.1:8032
Container: container_17797255561_0005_01_000003 on worker1_45087
LogAggregationType: AGGREGATED
=====
LogType: directory_info
LogLastModifiedTime: Mon May 25 19:14:54 +0000 2022
LogLength: 37158
LogContent:
ls -l
total 42
lrwxrwxrwx 1 root root 76 May 24 03:14 __spark_conf__ -> /tmp/hadoop-root/nm-local-dir/usercache/root/filescache/15/__/spark_conf___.zip
drwxr-xr-x 2 root root 16384 May 24 03:14 __spark_libs__
-rw-r--r-- 1 root root 89 May 24 03:14 container_tokens
-rw-r--r-- 1 root root 721 May 24 03:14 default_container_executor.sh
-rw-r--r-- 1 root root 468 May 24 03:14 default_container_executor_session.sh
-rw-r--r-- 1 root root 48877 May 24 03:14 launch_container.sh
lrwxrwxrwx 1 root root 77 May 24 03:14 py4j-0.10.9-src.zip -> /tmp/hadoop-root/nm-local-dir/usercache/root/filescache/15/py4j-0.10.9-src.zip
lrwxrwxrwx 1 root root 69 May 24 03:14 pyspark.zip -> /tmp/hadoop-root/nm-local-dir/usercache/root/filescache/15/pyspark.zip
drwxr-xr-x 2 root root 4096 May 24 03:14 tmp
Find -L -name *.jar
3615315 4 drwx-r--r-- 4 root root 4096 May 24 03:14 ./
3615328 48 -rw-r--r-- 1 root root 48877 May 24 03:14 ./launch_container.sh
3615329 16 drwxr-xr-x 2 root root 16384 May 24 03:14 ./__spark_libs__
3615408 1372 -r--r--r-- 1 root root 1408964 May 24 03:14 ./__spark_libs__/Jackson-databind-2.10.0.jar
3615099 296 -r--r--r-- 1 root root 382558 May 24 03:14 ./__spark_libs__/snakeyaml-1.24.jar
3615188 11632 -r--r--r-- 1 root root 1198731 May 24 03:14 ./__spark_libs__/suberretes-model-1.9.3.jar
3615128 64 -r--r--r-- 1 root root 64676 May 24 03:14 ./__spark_libs__/arrow-format-0.15.1.jar
3615192 196 -r--r--r-- 1 root root 288223 May 24 03:14 ./__spark_libs__/Ak2-api-2.4.1.jar
3615407 1264 -r--r--r-- 1 root root 1252426 May 24 03:14 ./__spark_libs__/levy-2.4.0.jar
3615137 132 -r--r--r-- 1 root root 134844 May 24 03:14 ./__spark_libs__/aircompressor-0.10.jar
3615099 1864 -r--r--r-- 1 root root 1906503 May 24 03:14 ./__spark_libs__/datanucleus-jdbc-4.1.10.jar
3615428 76 -r--r--r-- 1 root root 75930 May 24 03:14 ./__spark_libs__/spark-launcher-2.12-3.0.0.jar
3615406 188 -r--r--r-- 1 root root 194632 May 24 03:14 ./__spark_libs__/gsom-2.2.4.jar
3615427 44 -r--r--r-- 1 root root 43748 May 24 03:14 ./__spark_libs__/Jackson-module-parameters-2.10.0.jar
3615488 4988 -r--r--r-- 1 root root 502151 May 24 03:14 ./__spark_libs__/hadoop-hdfs-client-3.2.0.jar
3615474 184 -r--r--r-- 1 root root 181772 May 24 03:14 ./__spark_libs__/arrow-memory-0.15.1.jar
3615492 3288 -r--r--r-- 1 root root 3232888 May 24 03:14 ./__spark_libs__/hadoop-yarn-api-3.2.0.jar
361584 132 -r--r--r-- 1 root root 134876 May 24 03:14 ./__spark_libs__/breeze-metrics-2.12-1.0.jar
3615216 472 -r--r--r-- 1 root root 482484 May 24 03:14 ./__spark_libs__/jswales-core-2.12-3.4.0.jar
3615166 40 -r--r--r-- 1 root root 38332 May 24 03:14 ./__spark_libs__/hive-vector-code-gen-2.3.7.jar
3615272 656 -r--r--r-- 1 root root 648330 May 24 03:14 ./__spark_libs__/epire-4.0.jar
3615486 48892 -r--r--r-- 1 root root 41871642 May 24 03:14 ./__spark_libs__/hudi-spark3-bundle-2.12-0.10.0.jar
3615462 2832 -r--r--r-- 1 root root 2871680 May 24 03:14 ./__spark_libs__/spark-hive-thriftserver-2.12-3.0.0.jar
3615435 416 -r--r--r-- 1 root root 423175 May 24 03:14 ./__spark_libs__/okhttp-3.12.4.jar
3615481 832 -r--r--r-- 1 root root 847738 May 24 03:14 ./__spark_libs__/parquet-encoding-1.10.1.jar
3615171 64 -r--r--r-- 1 root root 43572 May 24 03:14 ./__spark_libs__/slf4j-api-1.7.30.jar
3615468 16 -r--r--r-- 1 root root 12984 May 24 03:14 ./__spark_libs__/hive-shims-scheduler-2.3.7.jar
3615387 144 -r--r--r-- 1 root root 147763 May 24 03:14 ./__spark_libs__/antlr-runtime-3.5.2.jar
3615208 40 -r--r--r-- 1 root root 48554 May 24 03:14 ./__spark_libs__/kerby-util-1.0.1.jar
3615079 80 -r--r--r-- 1 root root 79108 May 24 03:14 ./__spark_libs__/epire-metrics-2.12-1.0-17.0-M1.jar
3615239 224 -r--r--r-- 1 root root 224672 May 24 03:14 ./__spark_libs__/kerby-core-1.0.1.jar
3615483 184 -r--r--r-- 1 root root 185340 May 24 03:14 ./__spark_libs__/metrics-core-4.1.1.jar
3615482 28 -r--r--r-- 1 root root 27884 May 24 03:14 ./__spark_libs__/spark-avro-2.12-3.0.0.jar
3615473 4 -r--r--r-- 1 root root 4028 May 24 03:14 ./__spark_libs__/ahimsa-0.7.45.jar
3615469 348 -r--r--r-- 1 root root 344748 May 24 03:14 ./__spark_libs__/datanucleus-api-jdo-2.4.jar
3615438 984 -r--r--r-- 1 root root 1087582 May 24 03:14 ./__spark_libs__/mysql-connector-java-6.1.47.jar
3615465 1848 -r--r--r-- 1 root root 1914588 May 24 03:14 ./__spark_libs__/hadoop-yarn-server-common-3.2.0.jar
3615447 428 -r--r--r-- 1 root root 443863 May 24 03:14 ./__spark_libs__/joda-time-2.10.5.jar
3615474 324 -r--r--r-- 1 root root 338887 May 24 03:14 ./__spark_libs__/spark-yarn-2.12-3.0.0.jar
3615414 292 -r--r--r-- 1 root root 298460 May 24 03:14 ./__spark_libs__/spark-mesos-2.12-3.0.0.jar
3615245 12 -r--r--r-- 1 root root 12211 May 24 03:14 ./__spark_libs__/slf4j-log4j12-1.7.30.jar
3615386 36 -r--r--r-- 1 root root 33833 May 24 03:14 ./__spark_libs__/jersey-0.8.jar
3615175 64 -r--r--r-- 1 root root 42858 May 24 03:14 ./__spark_libs__/commons-logging-1.1.3.jar
3615488 120 -r--r--r-- 1 root root 139008 May 24 03:14 ./__spark_libs__/hadoop-hdfs-3.2.0.jar
3615462 56 -r--r--r-- 1 root root 54391 May 24 03:14 ./__spark_libs__/jobjenesis-2.5.1.jar
3615495 128 -r--r--r-- 1 root root 128414 May 24 03:14 ./__spark_libs__/r2j-1.1.jar

```

HBase Scan Verification

The metrics written to HBase were RMSE and R2 from the PySpark Linear Regression model. As previously stated, RMSE measures the average prediction error for GPA, and R2 measures how well the model explains GPA variation. These results were successfully stored in HBase, validating that the pipeline worked end-to-end.

```

bash-5.0# hbase shell
scan 'my_table'
2026-05-25 19:39:32,924 WARN [main] util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.3.6, r7414579f2620fca6b75146c29ab2726fc4643ac9, Wed Jul 28 22:24:42 UTC 2021
Took 0.0022 seconds
hbase(main):001:0> scan 'my_table'
ROW                                COLUMN+CELL
metrics1                           column=cf:r2, timestamp=2026-05-25T19:31:06.805, value=0.9778711163548931
metrics1                           column=cf:rmse, timestamp=2026-05-25T19:31:06.823, value=0.06562225809386474
1 row(s)
Took 0.8711 seconds
hbase(main):002:0>

```

Conclusion

The data was ingested via NiFi, loaded into HDFS, loaded into a Hive table, analyzed in Spark using SQL and MLlib, and the final evaluation metrics were written to HBase for easy querying. This type of pipeline, when applied to real-world data, can yield insightful, data-driven results that guide business decisions in ways not possible before the introduction of big data tools.